An independent cubic derivative singlet
Gram projection and the first spectral moments

Alan Day Chengheng*

Local research note — 6 November 2026

Abstract

The cubic fermion bilinear contains an exact derivative of the quadratic singlet. We isolate its Hermitian part, retain its third Gaussian chaos, and project out the cubic energy polynomial and the energy-weighted quadratic singlet. The remaining operator has strictly positive norm and nonzero dynamics in complex DSSYK double scaling. Its second and fourth frequency moments, and all nine two-arc moments through four external Hamiltonians, agree with a probe of crossing weight q^3. This gives a concrete candidate for the next independent singlet channel and a finite-degree target for an all-order kernel reduction.

1 Result and physical direction

Use standard CAR $\{c_i, c_j^\dagger\} = \delta_{ij}$, a Hermitian Hamiltonian $H = H_s$ with p creations and p annihilations in every normal-ordered monomial, and the full annealed trace $\langle Y\rangle = \mathbb{E}_J 2^{-N} \operatorname{Tr} Y$ at $\beta = \mu_{\text{chem}} = 0$. The complex Gaussian ensemble includes overlapping index sets. First take $N, p \to \infty$ at fixed

$$\lambda = p^2/N > 0, \qquad q = e^{-4\lambda}, \qquad \sigma^2 = \langle H^2\rangle = \mathcal{J}_s^2/\lambda.$$

Write $L = [H, \cdot\,]$, $A = \sum_i (Lc_i)^\dagger Lc_i$, $M_3 = \sum_i c_i^\dagger L^3 c_i$, and $T_3 = (M_3 + M_3^\dagger)/2$. Our operator is

$$R = T - aS - bZ, \quad (T, S, Z) = \mathcal{W}_3(T_3, \{H, A\}/2, H^3), \tag{1}$$

where $\mathcal{W}_3$ means third *Gaussian coupling chaos*, not fermion normal ordering. The coefficients are

$$a = -\frac{(1-q)^2}{1+q^2}, \qquad b = \frac{2\lambda(1-q)(1+q^3)}{(1+q^2)(1+q+q^2)}. \tag{2}$$

For $\widehat{R} = R/\sqrt{\langle R^2\rangle}$, the frequency moments are

$$\mu_2 = \langle \widehat{R} L^2 \widehat{R}\rangle = \frac{2\mathcal{J}_s^2}{\lambda}(1-q^3), \tag{3}$$

$$\mu_4 = \frac{2\mathcal{J}_s^4}{\lambda^2}(1-q)^2(1+q+q^2)(3q^3+6q^2+6q+5). \tag{4}$$

Thus $\mu_2 \to 24\mathcal{J}_s^2$ and $\mu_4 \to 1920\mathcal{J}_s^4$ as $\lambda \to 0^+$. These are microscopic frequency data for the Susskind singlet program [1]; they do not yet fix a bulk mass.

*AI research assistant (OpenAI).

2 Removing the exact time descendant

Define $B_{ab} = \sum_i (L^a c_i)^\dagger L^b c_i$. Since $(LY)^\dagger = -L(Y^\dagger)$, the product rule gives

$$LB_{ab} = -B_{a+1,b} + B_{a,b+1}. \tag{5}$$

Homogeneity in the fermions implies $B_{01} = -pH$, so $B_{02} = B_{11} = A$. Applying (5) twice gives

$$M_3 = 2B_{12} - B_{21}, \qquad M_3^\dagger = 2B_{21} - B_{12}, \tag{6}$$

$$T_3 = \tfrac{1}{2}(B_{12} + B_{21}), \qquad \boxed{M_3 = T_3 + \tfrac{3}{2}LA.} \tag{7}$$

These are exact finite-N identities for each realization. In real time $\partial_t^n c = i^n L^n c$; retaining these factors is essential when comparing the schematic derivative family in [1].

For a direct construction let $Q_f = \sum_i c_i^\dagger c_i$ and

$$\mathfrak{D}(Y) = Q_f Y - \sum_i c_i^\dagger Y c_i.$$

On a normal-ordered number-conserving monomial with k creation operators, $\mathfrak{D}$ multiplies by k. Expanding the commutators therefore gives

$$A = (2p - \mathfrak{D})H^2, \tag{8}$$

$$T_3 = (3p - \mathfrak{D})H^3 - \tfrac{3}{2}\{H, A\}. \tag{9}$$

This also explains the contraction marks used below. For three individually p-body factors V_1, V_2, V_3, set

$$K_{12} = ((2p - \mathfrak{D})(V_1 V_2))V_3, \tag{10}$$

$$K_{23} = V_1((2p - \mathfrak{D})(V_2 V_3)), \tag{11}$$

$$K_{13} = (3p - \mathfrak{D})(V_1 V_2 V_3) - K_{12} - K_{23}. \tag{12}$$

Each normal-order contraction reduces the total body rank by one. Consequently K_{ij} counts contractions between those two factors, including the nonadjacent pair 13. With all $V_i = H$,

$$T_3 = K_{13} - \tfrac{1}{2}(K_{12} + K_{23}), \qquad \{H, A\}/2 = \tfrac{1}{2}(K_{12} + K_{23}). \tag{13}$$

There is also an exact ensemble selection rule: $\langle T_3 L^n A \rangle = 0$ for every n. For even n, use $H \mapsto -H$. For odd n, the trace is purely imaginary by Hermiticity, but its average is real by complex conjugation of the couplings in the real occupation basis. The same argument applies to R and the previously constructed even quadratic singlet X. This removes the simple X time-descendant sector; energy-weighted composites still require the explicit projection (1).

3 A positive residual from the six-slot Gram form

For Gaussian coordinates J_α with covariance $C_{\alpha\beta}$,

$$\mathcal{W}_3(J_\alpha J_\beta J_\gamma) = J_\alpha J_\beta J_\gamma - C_{\alpha\beta}J_\gamma - C_{\alpha\gamma}J_\beta - C_{\beta\gamma}J_\alpha.$$

When computing a product of such insertions, Wick pairings internal to either three-slot insertion are excluded. This specifies the operator before taking double scaling. The lower, linear chaos of the raw T_3 has not been identified here with a conserved operator.

The Gram matrix $G_{ij} = \langle Y_i Y_j\rangle/\sigma^6$, for $Y = (T, S, Z)$, is the sum of the six cross pairings. Its independent entries are

$$G_{TT} = 2\lambda(4\lambda q^2 + 4\lambda q + q^2 - 4q + 3), \tag{14}$$

$$G_{TS} = -2\lambda(q-1)^2, \qquad G_{TZ} = -4\lambda q(q-1), \tag{15}$$

$$G_{SS} = 2\lambda(2\lambda q^3 + 2\lambda + q^2 + 1), \tag{16}$$

$$G_{SZ} = -2\lambda(q-1)(q^2+q+1), \tag{17}$$

$$G_{ZZ} = (q+1)(q^2+q+1). \tag{18}$$

Solving $\langle RS\rangle = \langle RZ\rangle = 0$ yields (2). Define

$$\mathcal{B} = 1 - q^3 + 4\lambda q^2(2q^2 - q + 2).$$

The Schur complement is

$$V := \frac{\langle R^2\rangle}{\sigma^6} = \frac{4\lambda(1+q)\mathcal{B}}{(1+q^2)(1+q+q^2)} = 32\lambda^2 - 96\lambda^3 + O(\lambda^4). \tag{19}$$

Since $0 < q < 1$ and $2q^2 - q + 2 > 0$, $V > 0$. In physical units $\langle R^2\rangle \sim 32\mathcal{J}_s^6/\lambda$. The direction therefore survives normalization even though its dimensionless Gram weight is small.

How the sums are evaluated. For a pairing involution π on ℓ Hamiltonian slots and two marks m_1, m_2, form the local entries

$$E_{uv} = \operatorname{sgn}(\pi(v) - u + \tfrac{1}{2}) - 2x\,\mathbf{1}_{(\pi(v),u)=m_1} - 2y\,\mathbf{1}_{(\pi(v),u)=m_2}.$$

Expand principal two-by-two minors with $x^2 = y^2 = 0$ and divide by their singleton signs. Adding the normalization contribution $-\binom{\ell}{2} - \ell/2$ gives $-4\,\mathrm{cr} + \alpha x + \beta y + \delta xy$. The fixed-degree double-scaling weight is consequently

$$q^{\mathrm{cr}}(\lambda\delta + \lambda^2\alpha\beta)$$

for two marks, $q^{\mathrm{cr}}\lambda\alpha$ for one, and q^{cr} for none. This follows from the Poisson pair-overlap limit; higher mode coincidences vanish at fixed degree. Equations (13) then give the displayed matrix. The nonadjacent mark is checked separately against finite CAR products, rather than inferred from an independent-probe formula.

4 Dynamics and a finite-degree kernel reduction

Define $K_{ij} = \langle (LY_i)^\dagger LY_j \rangle / \sigma^8$. Cyclic trace gives the computable real symmetric expression

$$\sigma^8 K_{ij} = \langle Y_i Y_j H^2 \rangle + \langle Y_j Y_i H^2 \rangle - 2\langle Y_i H Y_j H \rangle. \tag{20}$$

All Gaussian pairings are retained except those internal to a Wick-ordered insertion. There are 42 allowed pairings in each eight-slot arrangement, not just 4! fully cross pairings. The nonzero entries are

$$K_{TT} = 2\lambda(1-q^2)[4\lambda q^3 + 16\lambda q^2 + 4\lambda q + q^3 - 3q^2 - 3q + 5], \tag{21}$$

$$K_{TS} = 2\lambda(q-1)^3(q+1)(4\lambda q + q + 1), \tag{22}$$

$$K_{SS} = 2\lambda(1-q^2)(1+q^2)(4\lambda q + q + 1). \tag{23}$$

The Z row vanishes. For $v = (1, -a, -b)^T$ one obtains the particularly simple exact identity

$$v^T K v = 2(1-q^3)V, \qquad v^T K v = 768\lambda^3 + O(\lambda^4). \tag{24}$$

This proves (3) and nonzero dynamics at every $\lambda > 0$.

More information is available than this single ratio. Let

$$F_{uv} = \frac{\langle R H^u R H^v \rangle}{\langle R^2 \rangle \sigma^{u+v}}.$$

Exact marked enumeration gives the following complete set through four external Hamiltonians (with $F_{uv} = F_{vu}$):

$u+v$	(u,v)	F_{uv}
0	$(0,0)$	1
2	$(2,0)$	1
2	$(1,1)$	q^3
4	$(4,0)$	$2+q$
4	$(3,1)$	$q^3(2+q)$
4	$(2,2)$	$1+q^6+q^7$

All corresponding $\langle R H^u S H^v \rangle$ and $\langle R H^u Z H^v \rangle$ vanish. Combining the fourth-order row with the binomial expansion of L^4 gives (4).

These nine two-arc values coincide with the existing independent probe rule [2]: each Hamiltonian crossing of the probe carries $r = q^3$. This is a tested finite-degree reduction for a *coupling-dependent composite*; independence of its couplings has not been assumed. An all-order proof would permit use of the known q-Mehler two-point kernel. It would also predict $\langle \widehat{R}(t)\widehat{R}(0) \rangle \to \mathrm{sech}^6(2\mathcal{J}_s t)$ at fixed string time as $\lambda \to 0^+$. At this stage that full function is a conjecture supported by the computed moments.

5 Evidence, provenance, and the next calculation

The motivation is the derivative singlet family in Miyashita–Sekino–Susskind, Sec. 4.3 [1]. Their proposed $n = 3$ assignment motivates this test but does not supply the quantitative correlator. Berkooz–Narovlansky–Raj, Sec. 7 and Eqs. (7.1)–(7.3) [2], solve a probe whose random couplings are uncorrelated with those of H. We use their kernel as a comparison target; the new work here is the explicit same-coupling composite and its finite-degree reduction. No priority claim is made for the polynomial identities.

The saved checks comprise:

- Twelve operator identities in independent exact integer-complex SymPy CAR matrices for $(N, p) = (3, 1), (4, 2)$. The same identities also pass the direct occupation-basis code.
- 192 fixed six-slot words, including coherent repeated index sets, with all 16 choices of zero, adjacent, or nonadjacent marks: 3072 finite CAR/determinant comparisons.
- Six exact symbolic projection, positivity-form, kinetic, and small-λ certificates; nine two-arc equalities and 18 vanishing cross moments through degree four.

The analytic positivity follows from (19), rather than from a parameter scan. Double scaling is justified at fixed word degree; finite CAR checks test the contraction interpretation and signs, not convergence in N. The scripts record dependencies and input hashes. A parsing failure caused by the reserved word `lambda` was fixed before the higher-moment output was written; no earlier scientific output was replaced.

Next research step. Prove or disprove the q^3 reduction for arbitrary two-arc words. The useful target is cancellation of diagrams in which one of the three insertion slots contracts with an external H after the S, Z subtraction. A finite local endpoint identity, analogous to the earlier quadratic construction, could reduce the remaining three lines to one probe line. This is preferable to merely extending the moment table. In parallel with that analytic step, the linear Gaussian chaos of raw T_3 should be identified so that the relation to the unprojected derivative operator is complete.

The present positive residual establishes independence from the displayed cubic composites and from the simple X time-descendant sector in the ensemble inner product. A complete particle classification still requires the full operator dictionary and spatial dynamics. In particular, neither the derivative index nor the $r = q^3$ candidate label is by itself a mass eigenvalue. Local manuscript and PDF only; no Git commit, push, or archive was made in this stage.

References

[1] S. Miyashita, Y. Sekino and L. Susskind, *Holograms and Standard Models*, arXiv:2607.05678v2, Sec. 4.3, Eqs. (4.19)–(4.26).

[2] M. Berkooz, V. Narovlansky and H. Raj, *Complex Sachdev–Ye–Kitaev model in the double scaling limit*, arXiv:2006.13983v2, Sec. 7, Eqs. (7.1)–(7.3).